

Arduino-Based Flight Simulator Using Joystick

A hands-on engineering project demonstrating real-time aircraft control using an Arduino Uno and a 2-axis joystick module.

ENGINEERING PROJECT

EMBEDDED SYSTEMS



What Is a Flight Simulator?

A flight simulator replicates the experience of flying an aircraft in a controlled environment – used for pilot training, research, and entertainment.

Why It Matters

Simulators reduce training costs and risk while allowing safe practice of emergency scenarios.

Why Arduino?

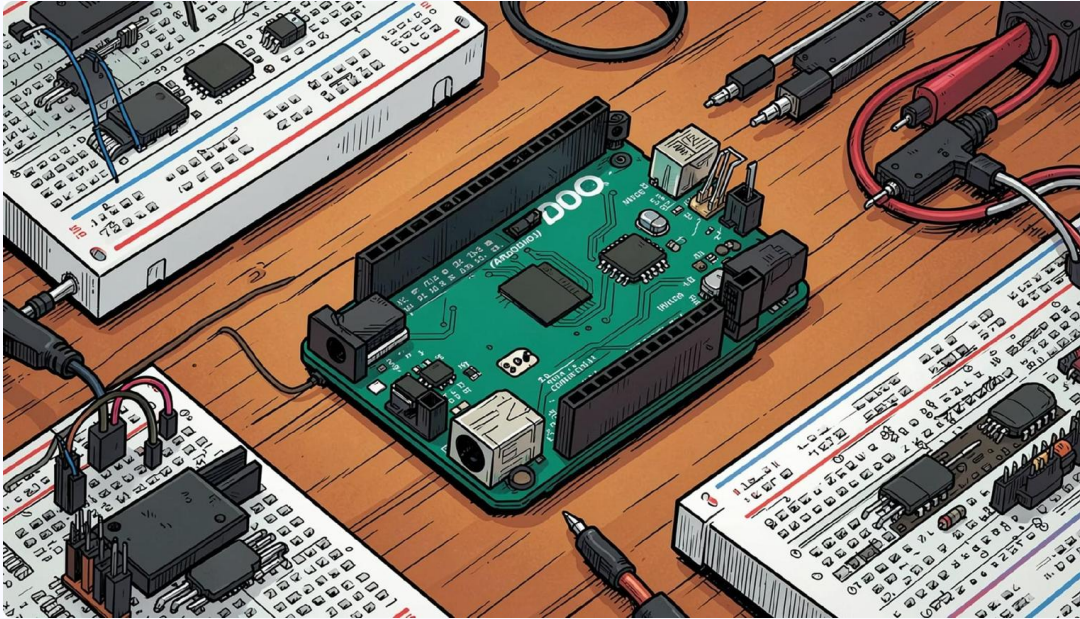
Arduino offers an affordable, open-source platform for building custom cockpit hardware and interfacing with simulation software.

DIY Approach

Students can replicate professional control systems at a fraction of the cost, gaining real embedded systems experience.



Project Objectives



What We Set Out to Achieve

- **Build a low-cost flight controller** using readily available components.
- **Learn Arduino programming** to read analog sensor inputs.
- **Interface hardware with simulation software** via USB serial communication.
- **Develop hands-on embedded systems skills** applicable to real-world engineering.

Components Required



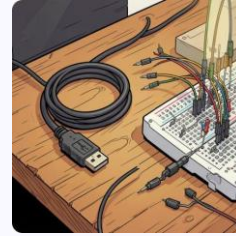
Arduino Uno

The brain of the system – reads joystick inputs and sends data to the PC.



2-Axis Joystick Module

Provides analog X and Y axis readings plus a push-button switch.



USB Cable & Wires

Connects the Arduino to the computer and links components on the breadboard.



Computer / Laptop

Runs the flight simulator software and receives serial data from the Arduino.

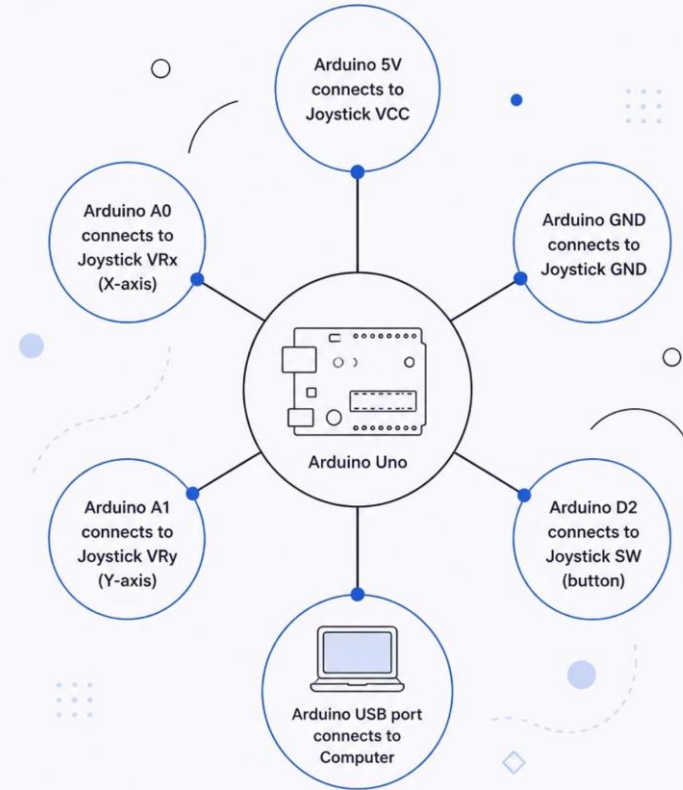
Hardware & Circuit Connections

Arduino Uno

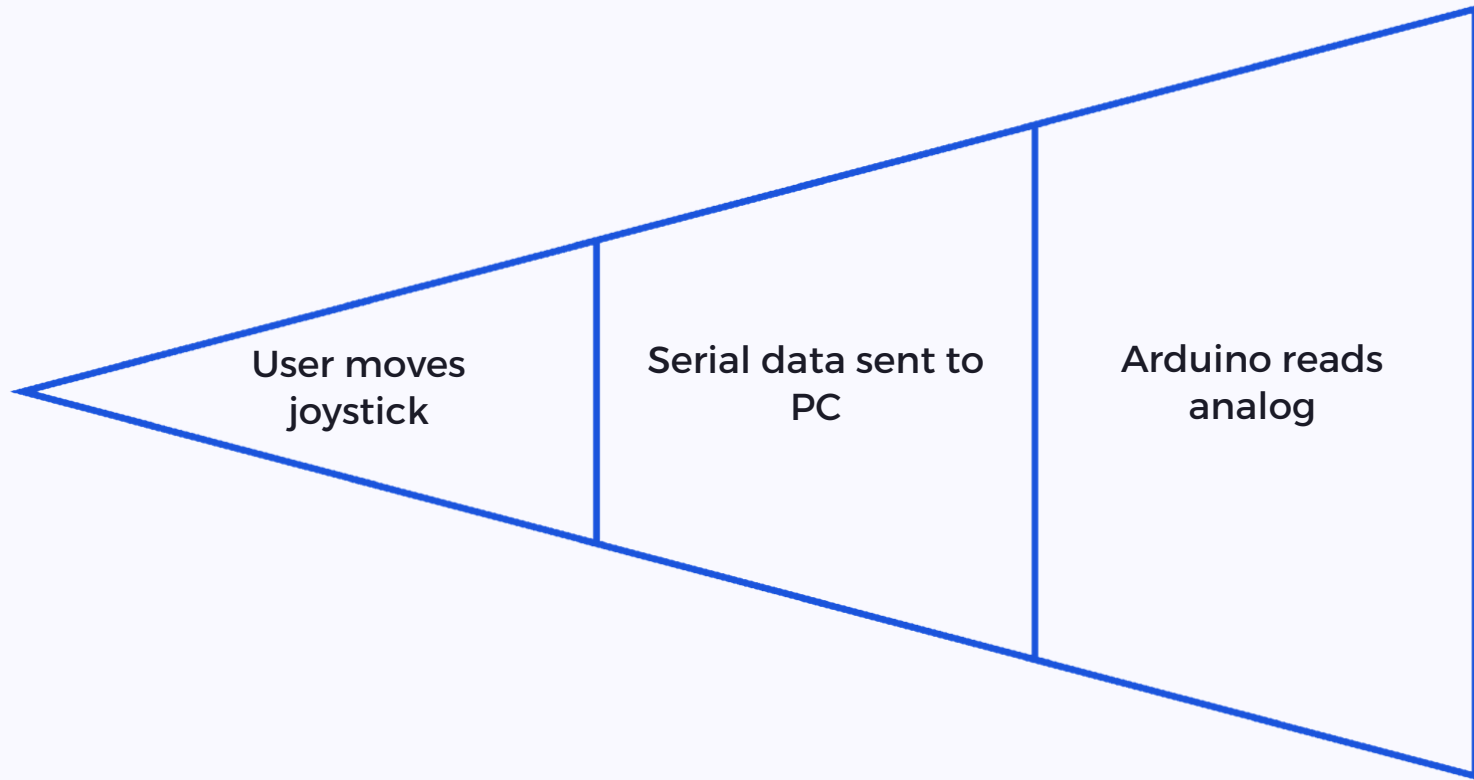
Based on the ATmega328P, the Uno features 14 digital I/O pins, 6 analog inputs, and a 16 MHz clock – ideal for reading joystick signals.

Joystick Module Pins

- **VCC** → 5V power supply
- **GND** → Ground
- **VRx** → Analog pin A0 (X-axis)
- **VRy** → Analog pin A1 (Y-axis)
- **SW** → Digital pin D2 (button)

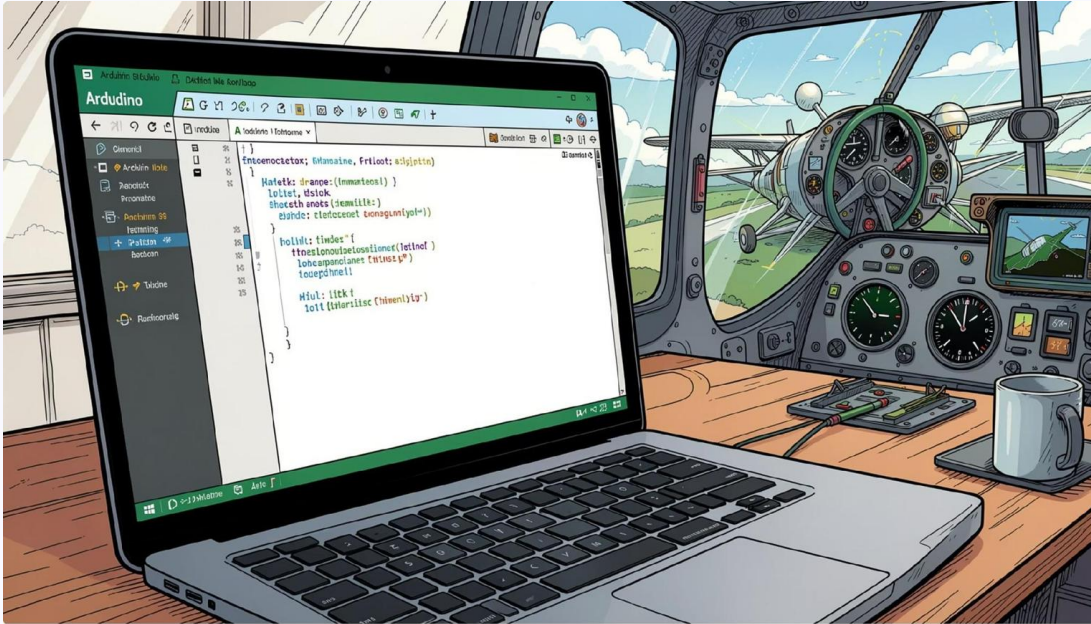


Working Principle



The joystick outputs analog voltages that the Arduino converts to digital values. These values are mapped to aircraft control commands and transmitted over USB in real time, creating a responsive flight control experience.

Software & Program Flow



Software Stack

Arduino IDE

Write, compile, and upload code to the Arduino board.

Joystick Library

Maps analog inputs to HID joystick commands (optional).

FlightGear / X-Plane

Open-source simulators that accept serial or joystick input.

Program Flow

1

Read analog values from pins A0 and A1

2

Map values from 0-1023 to control range

3

Send serial data over USB to PC

4

Simulator updates aircraft pitch and roll in real time



Live Demonstration

✈ Pitch Control

Pushing the joystick forward tilts the aircraft nose down; pulling back raises it./...//

1. ↔ Roll Control
2. Moving left or right banks the aircraft, simulating aileron input.

⚡ Real-Time Response

Changes are reflected in the simulator with minimal latency via USB serial.

Advantages & Limitations

✓ Advantages

- **Low cost** – built with components under \$20
- **Easy to assemble** – minimal soldering required
- **Educational value** – covers electronics, coding, and aviation
- **Real-time control** – responsive aircraft movement
- **Expandable** – add throttles, switches, and pedals

⚠ Limitations

- **Basic controls only** – pitch and roll, no full 6-axis
- **No force feedback** – joystick lacks resistance cues
- **Limited precision** – analog noise affects accuracy
- **Software dependency** – requires a PC and simulator

Future Scope & Conclusion



Throttle & Rudder

Add potentiometer-based throttle and rudder pedals for full 3-axis control.



Cockpit Instruments

Drive LCD displays and LED indicators to replicate real cockpit gauges.



VR Integration

Combine with a VR headset for an immersive, head-tracked flight experience.



Conclusion: This project demonstrates how Arduino and a simple joystick can create an affordable, interactive flight simulator – bridging embedded systems, software, and aviation in a single hands-on build.

Thank You

Questions & Answers

